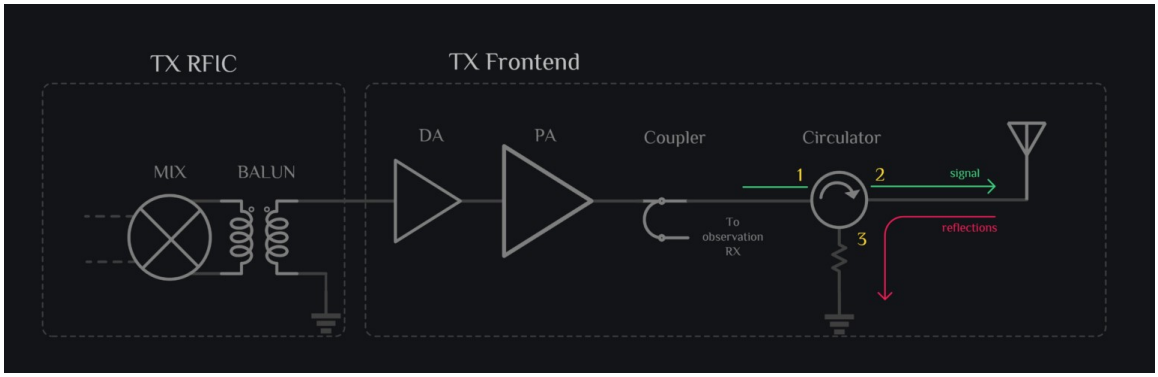
Loadline Design

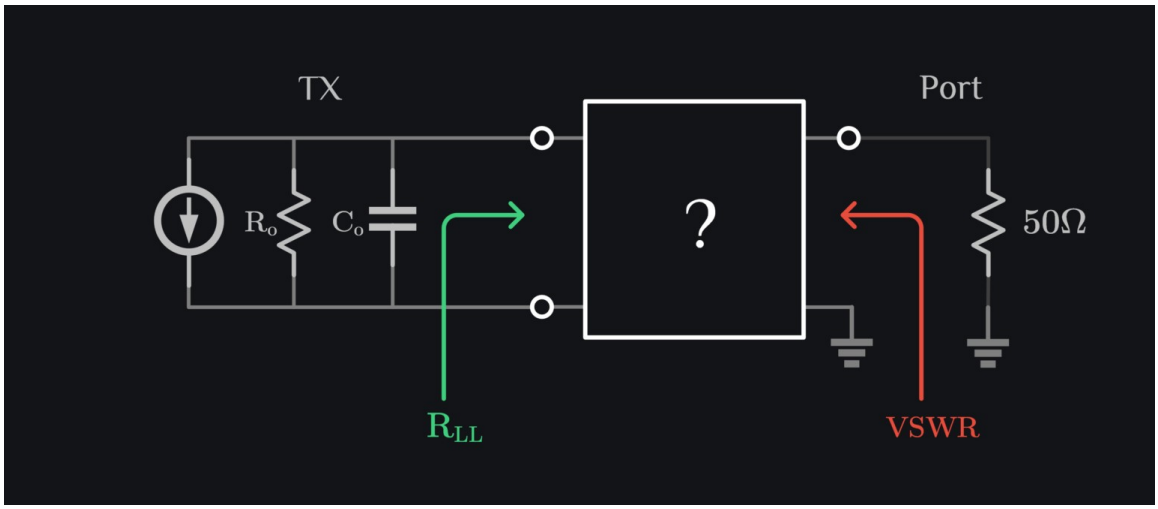
The output of a power amplifier (PA) is terminated with a specific impedance. This impedance is required for optimal linearity or power generation or efficiency or combination of such specs, and is determined by loadpull analysis. We call it loadline (https://www.youtube.com/watch?v=FsiCUr6tZBs). While a loadline helps PA deliver its performance, the matching (VSWR) gets ruined because we do not present conjugate load to PA. Typically, we insert a circulator between antenna and PA such that reflections don't reach back to PA, and get absorbed in port 3 of circulator as shown in image below.



This works fine for front-end modules where off-the-shelves blocks can be patched together to curb the ailment. In context of TX RFIC where adding an circulator might not be possible (as they are made of ferrite), we discuss how loadline can be designed while also meeting the VSWR requirements.

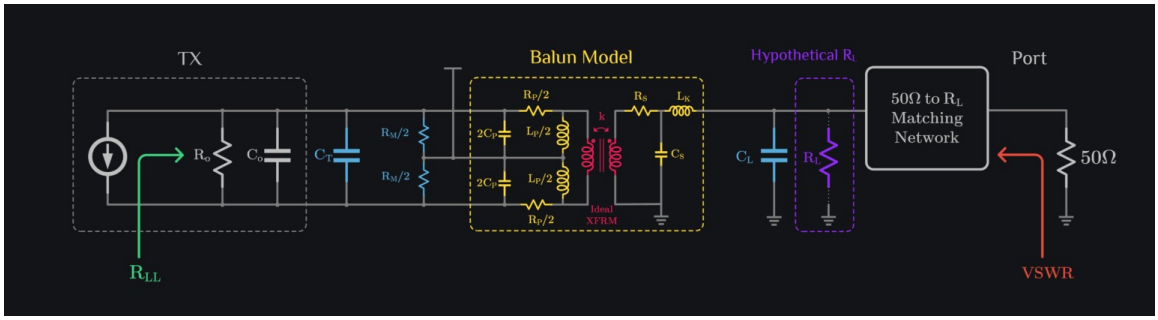
Problem Statement

Say we have a TX with output resistance R_o and output capacitance C_o. We have a requirement to present a certain resistance R_{LL} to this TX, and also maintain certain VSWR looking from 50Ω port. TX is differential, so we also want to convert it to single ended while doing so. Our job is to figure out how should the white box be designed.



Solution

One of the ways we can accomplish this is shown in image below. First, we add a balun to convert differential to single ended. Second, we add a resistor R_M to introduce loss into system as we will see later that simultaneous R_{LL} and VSWR requirements cannot be satisfied without adding R_M (also see a note on similar topic in appendix). We also add a capacitor C_T at primary of balun to tune out primary inductance partially, and C_L at secondary to tune out leftover inductance. (We mentioned tuning out primary inductance but not secondary – because there is only one inductance to tune out, the other inductance is a leakage inductance L_K which is very small or zero if coupling coefficient is one, see transformer model # 4 (https://www.rfinsights.com/insights/design/concepts/rfic-transformer-model/)). One can just use C_T or C_L to tune out, we distributed the cap because it gives us a knob for loss optimization (more on this later). Our goal is to find R_L and C_L such that both R_{LL} and VSWR are satisfied. Note R_L is not a physical resistor, it the resistance value to which we want our 50Ω port to convert to.



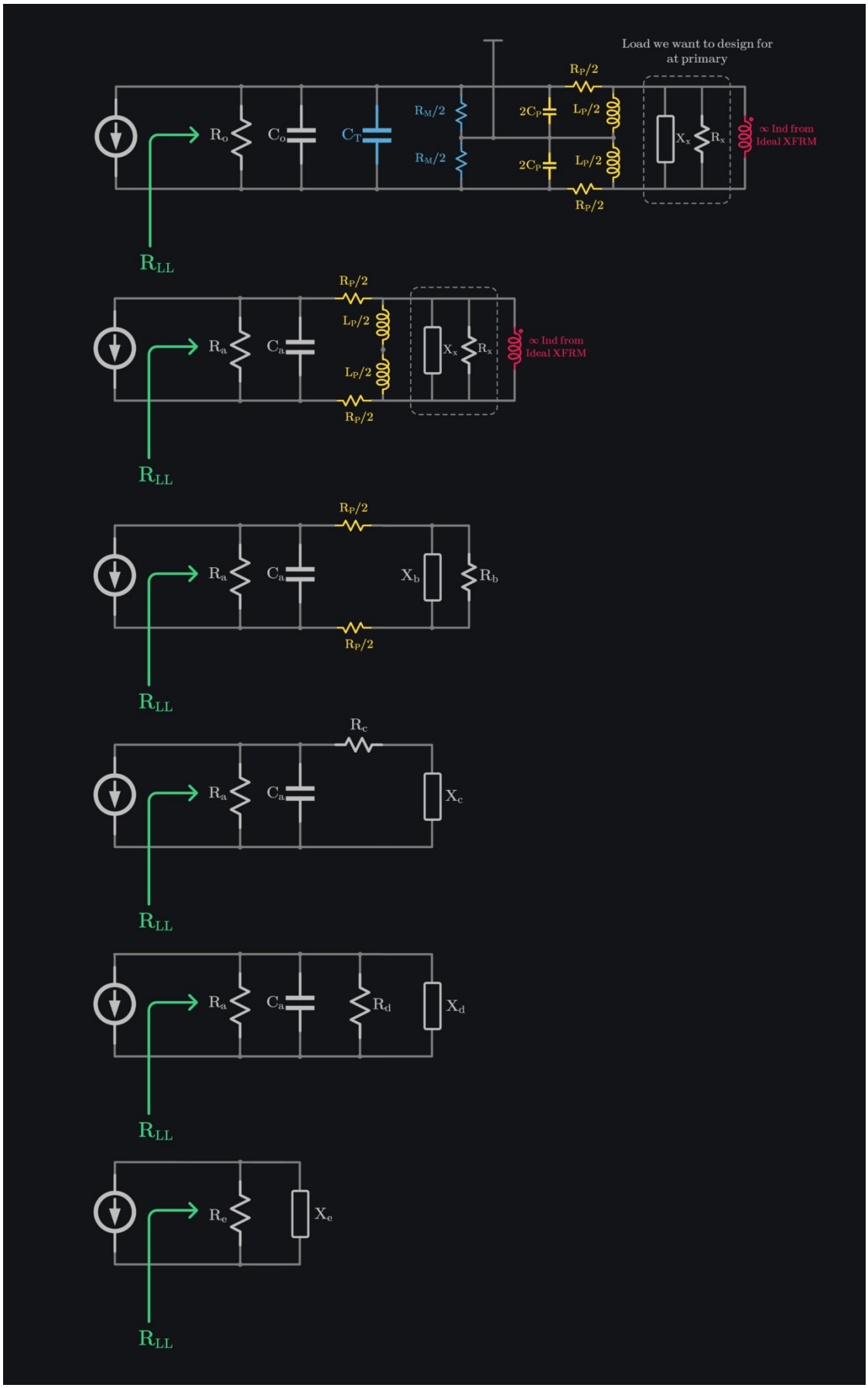
Loadline Design

Primary Side Calculations

Let's put some numbers in. Say we have 5kΩ R_o and 100fF C_o. We have C_T, R_M, L_p, L_s, k, Q_p, Q_s at our disposal to optimize. We start with 5nH L_p. It's just a guess, you can use any L_p, things you would need to consider in choosing L_p are:

- how much Q_p are you getting (the higher the better)
- what is the self-resonance frequency (the higher the better)
- how much area it takes (the lower the better)
- tuning capability (a higher L_p would need small tuning cap, which makes design very sensitive to parasitic and process variations)
- how much k you get (the higher the better)

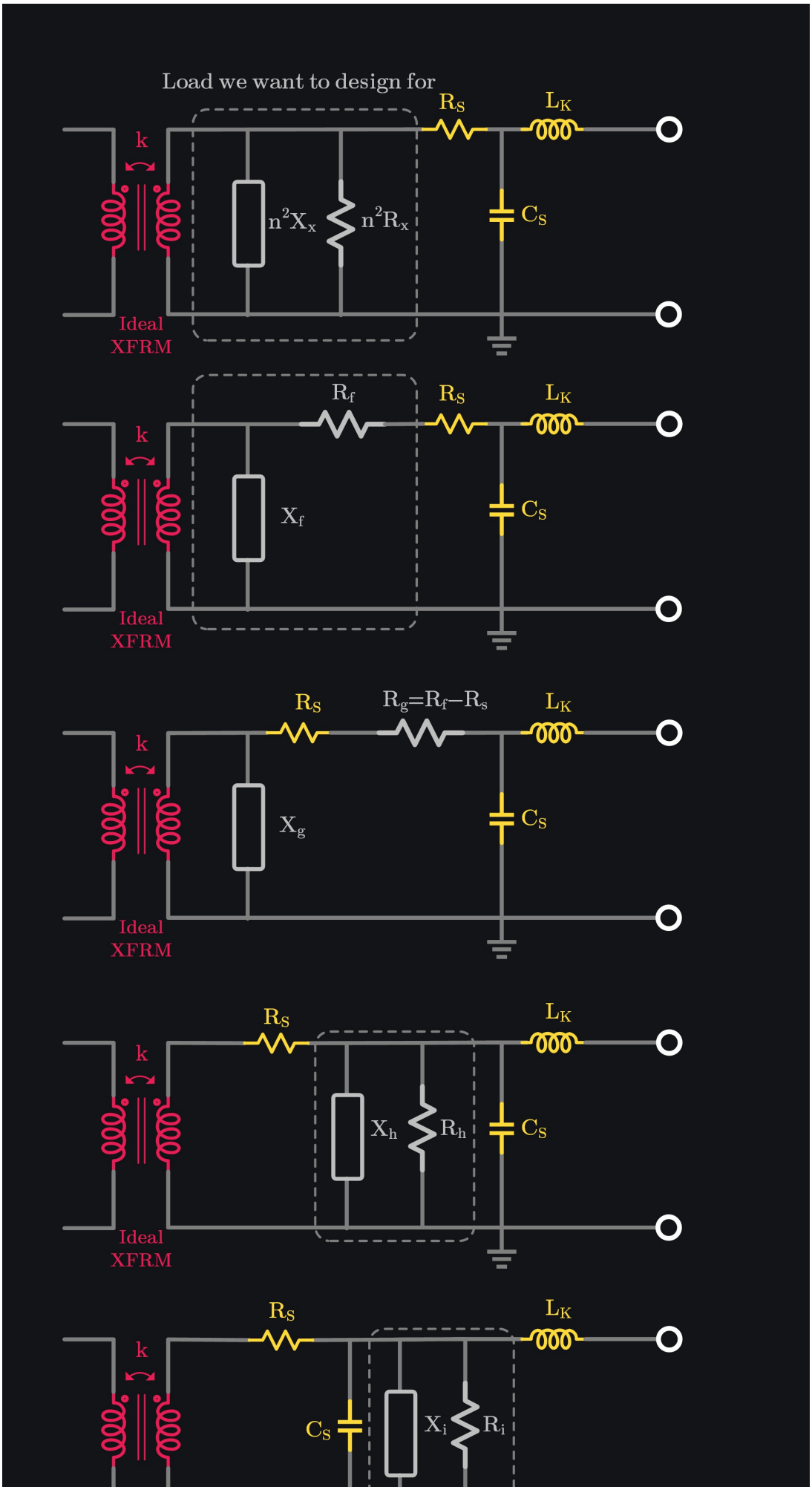
L_s is usually chosen to be 0.707 or 1.414 of L_p because this aspect ratio usually gives the highest k. Let's make R_M infinite for now, and choose a random value of C_T, and begin! First, let us find what values of resistance and reactance do we need looking into primary of balun, that would get us desired loadline. Let's call them R_x and X_x. We believe an analytical solution maybe worked out but we went with iterative solution in Microsoft Excel. We assume some initial value of R_x and C_x, and start simplifying the circuit till we get to R_e and X_e. This is shown in image to the right starting from top and ending at bottom. Excel would keep iterating different values of R_x and X_x until it gets us $R_e = R_{LL}$ and $X_e = \infty$, which is what we would want to see pure R_{LL}.

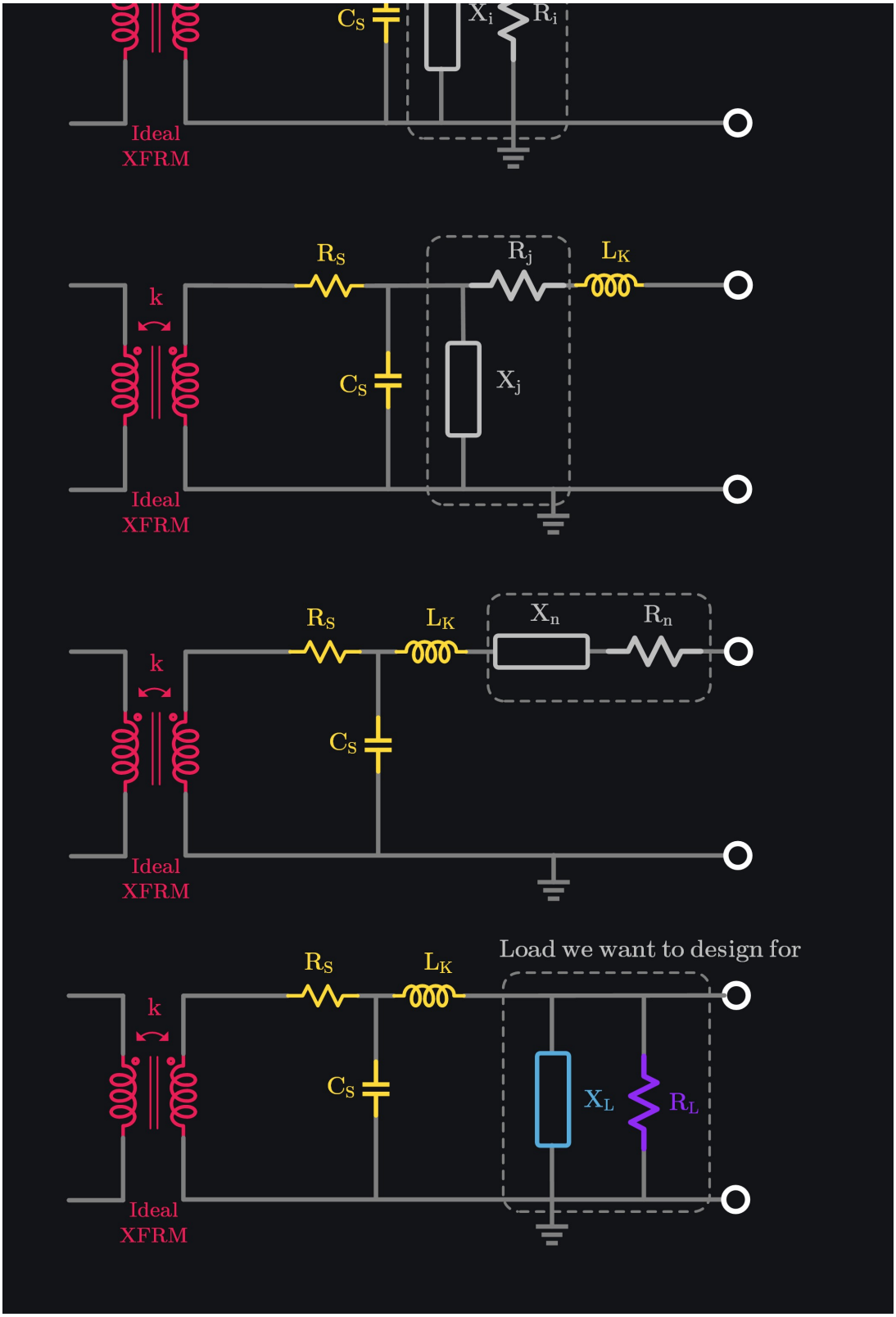


Secondary Side Calculations

Once we have found R_x and X_x needed at primary, we can multiply it by turn ratio square and figure out an equivalent RX needed at secondary. We then move this RX step by step towards the output, and finally arrive at R_L and C_L values. Basically this would mean: attach a C_L capacitor at output node, and attach a matching network that transforms port 50Ω to R_L. If done, your R_L C_L would transform through balun parasitics and other components we added, and once they make it to TX current source, all the imaginary part would have been stripped off and you should see real R_{LL}.

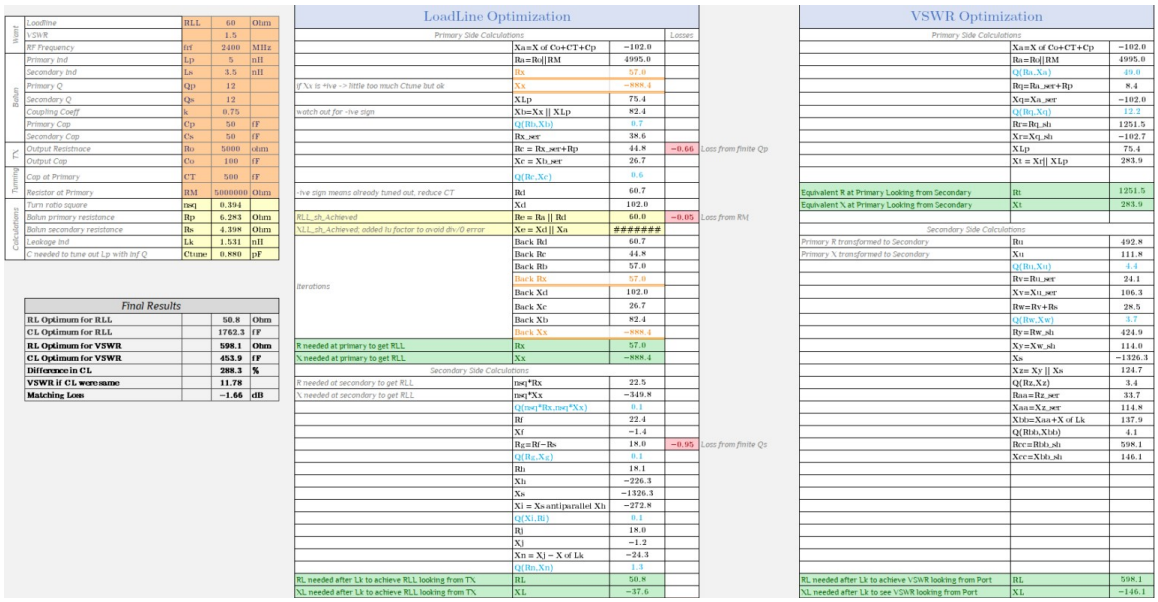
(Note it is also possible that your matching network transforms port 50Ω to R_L || X_L directly, so one can remove C_L)


Load we want to design for
R_S
L_K
k
n^2X_x
n^2R_x
C_S
Ideal
XFRM
R_f
X_f
$R_g=R_f-R_s$
X_g
X_h
R_h
C_S
X_i
R_i



We built an Excel calculator which details these calculations. Different parameters we assumed are in oranges boxes and R_L C_L values under Final Results. Our goal was 60Ω R_{LL}, and our calculator says we need to have $R_L = 50.8\Omega$ and $C_L = 1762.3fF$ to achieve this R_{LL}. Ok great, how about VSWR though? Our Excel sheet does similar calculations for VSWR and shows that we need $R_L = 598.1\Omega$ and $C_L = 453.9fF$ to get 1:1 VSWR. This R_L C_L is way

way off from what we need for R_{LL}. This is not surprising as VSWR is simplify looking for conjugate load which happens to be very different from our desired R_{LL}. In other words, if we had just gone for conjugate match for TX output, our VSWR would have been perfect but we would not have gotten the R_{LL} we wanted.



Input	Loadline	RLL	60	Ohm
	VSWR		1.5	
	RF Frequency	frf	2400	MHz
Balun	Primary Ind	Lp	5	nH
	Secondary Ind	Ls	3.5	nH
	Primary Q	Qp	12	
	Secondary Q	Qs	12	
	Coupling Coeff	k	0.75	
	Primary Cap	Cp	50	fF
	Secondary Cap	Cs	50	fF
TX	Output Resistnace	Ro	5000	ohm
	Output Cap	Co	100	fF
Primary	Cap at Primary	CT	500	fF
	Resistor at Primary	RM	5000000	Ohm
Calculations	Turn ratio square	nsq	0.394	
	Balun primary resistance	Rp	6.283	Ohm
	Balun secondary resistance	Rs	4.398	Ohm
	Leakage Ind	Lk	1.531	nH
	C needed to tune out Lp with Inf Q	Ctune	0.880	pF

Final Results			
RL Optimum for RLL		**50.8**	**Ohm**
CL Optimum for RLL		**1762.3**	**fF**
RL Optimum for VSWR		**598.1**	**Ohm**
CL Optimum for VSWR		**453.9**	**fF**
Difference in CL		**288.3**	**%**
VSWR if CL were same		**11.78**	
Matching Loss		**−1.66**	**dB**

LoadLine Optimization

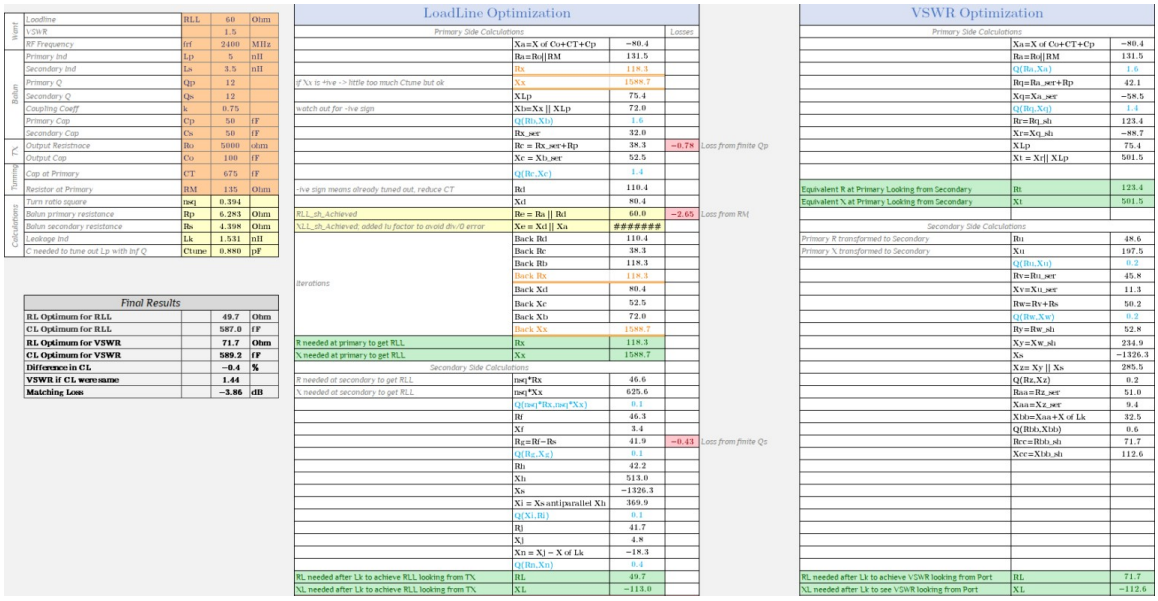
Primary Side Calculations			Losses
	Xa=X of Co+CT+Cp	−102.0	
	Ra=Rs||RM	4995.0	
	Rx	57.0	
if Xx is +ive -> little too much Ctune but ok	Xx	−888.4	
	XLp	75.4	
watch out for -ive sign	Xb=Xx || XLp	82.4	
	Q(Rb,Xb)	0.7	
	Rx_ser	38.6	
	Rc = Rx_ser+Rp	44.8	−0.66 Loss from finite Qp
	Xc = Xb_ser	26.7	
	Q(Rc,Xc)	0.6	
-ive sign means already tuned out, reduce CT	Rt	60.7	
	Xd	102.0	
RLL_sh_Achieved	Re = Ra || Rt	60.0	−0.05 Loss from RM
XLL_sh_Achieved: added to factor to avoid div/0 error	Xe = Xd || Xa	#######	
Iterations	Back Rt	60.7	
	Back Re	44.8	
	Back Rb	57.0	
	Back Rx	57.0	
	Back Xd	102.0	
	Back Xc	26.7	
	Back Xb	82.4	
	Back Xx	−888.4	
R needed at primary to get RLL	Rx	57.0	
X needed at primary to get RLL	Xx	−888.4	
Secondary Side Calculations			
R needed at secondary to get RLL	nsq*Rx	22.5	
X needed at secondary to get RLL	nsq*Xx	−349.8	
	Q(nsq*Rx,nsq*Xx)	0.1	
	Rf	22.4	
	Xf	−1.4	
	Rg=Rf−Rs	18.0	−0.95 Loss from finite Qs
	Q(Rg,Xg)	0.1	
	Rh	18.1	
	Xh	−226.3	
	Xs	−1326.3	
	Xi = Xs anti|parallel Xh	−272.8	
	Q(Xi,Ri)	0.1	
	Rj	18.0	
	Xj	−1.2	
	Xn = Xj − X of Lk	−24.3	
	Q(Rn,Xn)	1.3	
RL needed after Lk to achieve RLL looking from TX	RL	50.8	
XL needed after Lk to achieve RLL looking from TX	XL	−37.6	

VSWR Optimization

Primary Side Calculations		
	Xa=X of Co+CT+Cp	−102.0
	Ra=Rs||RM	4995.0
	Q(Ra,Xa)	49.0
	Rq=Ra_ser+Rp	8.4
	Xq=Xa_ser	−102.0
	Q(Rq,Xq)	12.2
	Rr=Rq_sh	1251.5
	Xr=Xq_sh	−102.7
	XLp	75.4
	Xt = Xr|| XLp	283.9
Equivalent R at Primary Looking from Secondary	Rt	1251.5
Equivalent X at Primary Looking from Secondary	Xt	283.9
Secondary Side Calculations		
Primary R transformed to Secondary	Rv	492.8
Primary X transformed to Secondary	Xv	111.8
	Q(Rv,Xv)	4.4
	Rv=Rv_ser	24.1
	Xv=Xv_ser	106.3
	Rw=Rv+Rs	28.5
	Q(Rw,Xw)	3.7
	Ry=Rw_sh	424.9
	Xy=Xw_sh	114.0
	Xs	−1326.3
	Xz= Xy || Xs	124.7
	Q(Rz,Xz)	3.4
	Rza=Rz_ser	33.7
	Xza=Xz_ser	114.8
	Xtb=Xza+X of Lk	137.9
	Q(Rtb,Xtb)	4.1
	Rrc=Rtb_sh	598.1
	Xrc=Xtb_sh	146.1
RL needed after Lk to achieve VSWR looking from Port	RL	598.1
XL needed after Lk to see VSWR looking from Port	XL	−146.1

Simultaneous Loadline and VSWR Optimization

How can we remedy this situation where we have different R_L C_L values to meet given R_{LL} and VSWR? By adding R_M. See that our loss in image above is 1.66dB, once we add R_M we can tradeoff loss with VSWR. Let's set our VSWR target to 1:1.5. By playing with R_M and C_T (of course you can play with other variables too, everything is on the table) we can arrive at R_L C_L values which will give the R_{LL} we want while meeting the VSWR. Image below shows that we don't need different value of C_L now, and R_L are still somewhat different but they meet VSWR condition. You can make R_L values similar too by reducing R_M at the expense of more loss. We are already hitting 3.86dB loss to get 1.5 VSWR. Our finally R_L and C_L are 49.7Ω and 587fF.



Input				
	Loadline	RLL	60	Ohm
	VSWR		1.5	
	RF Frequency	frf	2400	MHz
Balun	Primary Ind	Lp	5	nH
	Secondary Ind	Ls	3.5	nH
	Primary Q	Qp	12	
	Secondary Q	Qs	12	
	Coupling Coeff	k	0.75	
	Primary Cap	Cp	50	fF
	Secondary Cap	Cs	50	fF
CT	Output Resistnace	Ro	5000	ohm
	Output Cap	Co	100	fF
Primary	Cap at Primary	CT	675	fF
	Resistor at Primary	RM	135	Ohm
	Turn ratio square	nsq	0.394	
Calculations	Balun primary resistance	Rp	6.283	Ohm
	Balun secondary resistance	Rs	4.398	Ohm
	Leakage Ind	Lk	1.531	nH
	C needed to tune out Lp with Inf Q	Ctune	0.880	pF

Final Results			
RL Optimum for RLL		**49.7**	**Ohm**
CL Optimum for RLL		**587.0**	**fF**
RL Optimum for VSWR		**71.7**	**Ohm**
CL Optimum for VSWR		**589.2**	**fF**
Difference in CL		**−0.4**	**%**
VSWR if CL were same		**1.44**	
Matching Loss		**−3.86**	**dB**

LoadLine Optimization

Primary Side Calculations			Losses
	Xa=X of Co+CT+Cp	−80.4	
	Ra=Ro||RM	131.5	
	Rx	118.3	
if Xx is +ive -> little too much Ctune but ok	Xx	1588.7	
	XLp	75.4	
watch out for -ive sign	Xb=Xx || XLp	72.0	
	Q(Rb,Xb)	1.6	
	Rx_ser	32.0	
	Rc = Rx_ser+Rp	38.3	−0.78 Loss from finite Qp
	Xc = Xb_ser	52.5	
	Q(Rc,Xc)	1.4	
-ive sign means already tuned out, reduce CT	Rt	110.4	
	Xt	80.4	
RLL_sh_Achieved	Re = Ra || Rt	60.0	−2.65 Loss from RM
XLL_sh_Achieved; added tu foctor to avoid div/0 error	Xe = Xt || Xa	#######	
Iterations	Back Rt	110.4	
	Back Rc	38.3	
	Back Rb	118.3	
	Back Rx	118.3	
	Back Xt	80.4	
	Back Xc	52.5	
	Back Xb	72.0	
	Back Xx	1588.7	
R needed at primary to get RLL	Rx	118.3	
X needed at primary to get RLL	Xx	1588.7	
Secondary Side Calculations			
R needed at secondary to get RLL	nsq*Rx	46.6	
X needed at secondary to get RLL	nsq*Xx	625.6	
	Q(nsq*Rx,nsq*Xx)	0.1	
	Rf	46.3	
	Xf	3.4	
	Rg=Rf−Rs	41.9	−0.43 Loss from finite Qs
	Q(Rg,Xg)	0.1	
	Rb	42.2	
	Xb	513.0	
	Xs	−1326.3	
	Xi = Xs antiparallel Xb	369.9	
	Q(Xi,Ri)	0.1	
	Rj	41.7	
	Xj	4.8	
	Xn = Xj − X of Lk	−18.3	
	Q(Rn,Xn)	0.4	
RL needed after Lk to achieve RLL looking from TX	RL	49.7	
XL needed after Lk to achieve RLL looking from TX	XL	−113.0	

VSWR Optimization

Primary Side Calculations		
	Xa=X of Co+CT+Cp	−80.4
	Ra=Ro||RM	131.5
	Q(Ra,Xa)	1.6
	Rq=Ra_ser+Rp	42.1
	Xq=Xa_ser	−58.5
	Q(Rq,Xq)	1.4
	Rr=Rq_sh	123.4
	Xr=Xq_sh	−88.7
	XLp	75.4
	Xt = Xr|| XLp	501.5
Equivalent R at Primary Looking from Secondary	Rt	123.4
Equivalent X at Primary Looking from Secondary	Xt	501.5
Secondary Side Calculations		
Primary R transformed to Secondary	Rtt	48.6
Primary X transformed to Secondary	Xtt	197.5
	Q(Rtt,Xtt)	0.2
	Rv=Rtt_ser	45.8
	Xv=Xtt_ser	11.3
	Rw=Rv+Rs	50.2
	Q(Rw,Xw)	0.2
	Ry=Rw_sh	52.8
	Xy=Xw_sh	234.9
	Xs	−1326.3
	Xz= Xy || Xs	285.5
	Q(Rz,Xz)	0.2
	Raa=Rz_ser	51.0
	Xaa=Xz_ser	9.4
	Xtb=Xaa+X of Lk	32.5
	Q(Rtb,Xtb)	0.6
	Rrc=Rtb_sh	71.7
	Xrc=Xtb_sh	112.6
RL needed after Lk to achieve VSWR looking from Port	RL	71.7
XL needed after Lk to see VSWR looking from Port	XL	−112.6

Now one might ask that role of R_M is clear, but what do we need C_T for. Can we get rid of C_T and just use C_L to tune out balun? The answer is yes but at the expense of slightly higher loss (not always though, devil lies in details, one need to analyze the particular situation). C_T was placed strategically. Ideally, we should tune out things where they emerge, meaning we should have tuned L_p right where it was (at primary). If we don't tune it here, and instead try to get the -ive reactive part all the way from C_L, it will go through impedance transformations at different nodes (as shown in images above) and at some node if Q is high, it will result in higher loss. How do you know at which node Q is high? You can lookup the Q of all the nodes in our Excel calculator, it's colored sky blue.

Another question one might ask is if loss gets us VSWR, can we get rid of R_M and instead make our balun intentionally lossy (i.e., lower Q)? The answer is not really. It turns out series and shunt losses are different. A series resistor creates two nodes in the circuit and hence creates asymmetry in left and right impedances. For example, if you see that optimum of VSWR and R_{LL} don't align in frequency that is maybe you got optimum R_{LL} at 2.4GHz but optimum VSWR at 2.5GHz, know that this is because of series resistor somewhere. A shunt resistor keeps this symmetry. Since inductor's loss is physically a series resistor, we always want to maximize Q to reduce this resistor. Also there are other benefits of putting R_M (as shown in our circuit, center of R_M goes to supply, we intentionally made it a common mode), this helps in providing even harmonics a path to ground. Otherwise, they will be forced to find a way to circulate in your circuit and create non-linearities you wouldn't have imagined (like weird combinations of mixing products with your signal, that will land back at your IM3s/5s for example).

Loadline Design Verification in Cadence

Let's now head to Cadence and plug our numbers in to see if our methodology works. Cadence schematic is shown below. Note that there is no explicit L_K here because that was just for our math, real L_K is modelled by finite k which we have added in our circuit using "mind" instance in analogLib. Also note that there are two circuits shown but they are exactly the same, just different excitations: one measures R_{LL} and other measures VSWR.

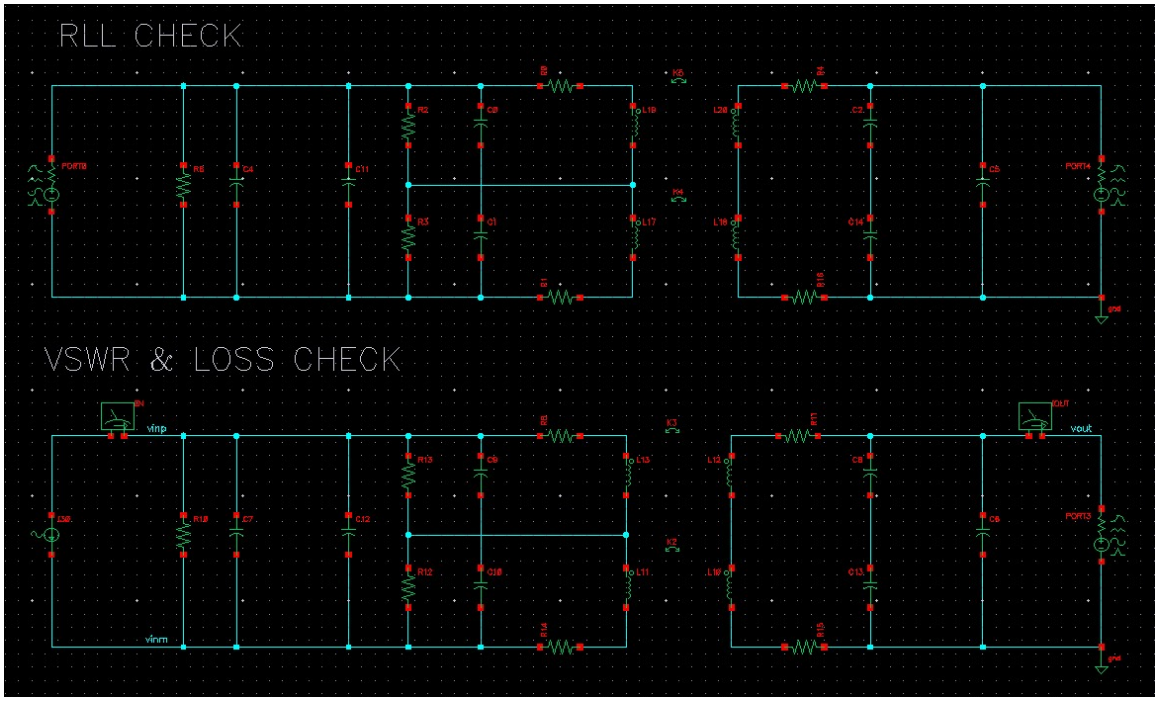
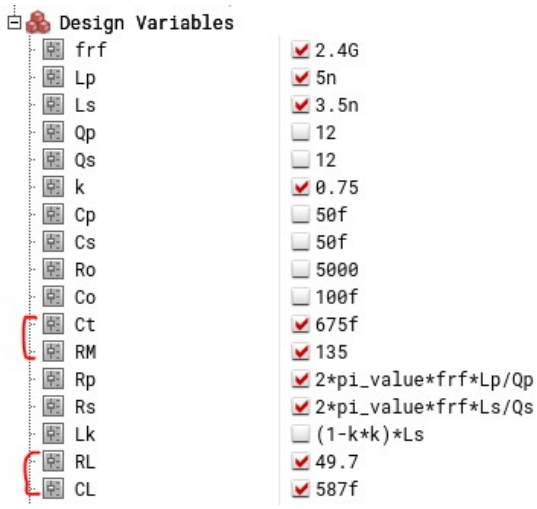
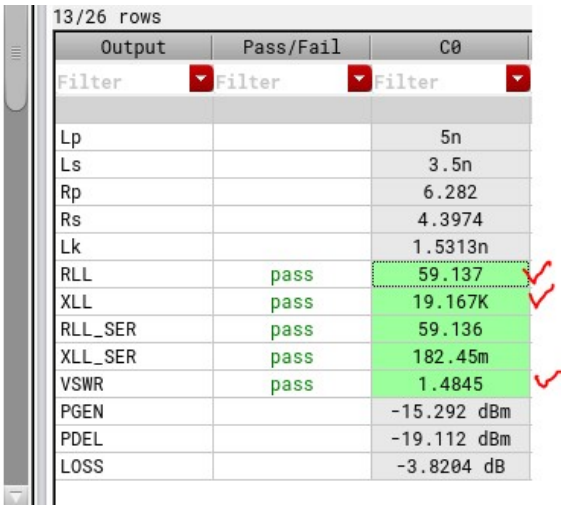


Image below shows simulation results. Our target was 60Ω R_{LL} and 1:1.5 VSWR. R_{LL}, VSWR and loss are match pretty well with our calculations.





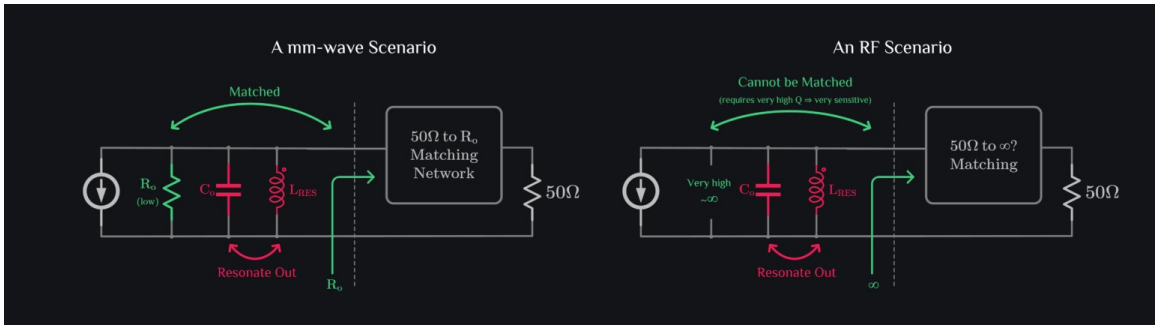
Output	Pass/Fail	C0
Filter	Filter	Filter
Lp		5n
Ls		3.5n
Rp		6.282
Rs		4.3974
Lk		1.5313n
RLL	pass	59.137
XLL	pass	19.167K
RLL_SER	pass	59.136
XLL_SER	pass	182.45m
VSWR	pass	1.4845
PGEN		-15.292 dBm
PDEL		-19.112 dBm
LOSS		-3.8204 dB

Excel calculator is attached below. One can start with this calculator to gain insights into circuit (like where is loss coming from, what variable does what, and which variable needs tight control in terms of design sensitivity) and later move on to directly optimize in Cadence.

Loadline Calculator (https://www.rfinsights.com/wp-content/uploads/2023/06/LoadlineCalc.xlsx)

Appendix

A subtle difference between mm-wave and RF impedance matching: at mm-wave, transistor output already contains low resistive part (either with existing losses in the circuit or through numerous feedback paths between output and different nodes of circuit that eventually introduce real part in output impedance), therefore one can just go on matching port to transistor output impedance. At RF, real part of output impedance is quite high (almost an open circuit) making it impossible to match to 50\(\Omega\) because matching network requires very high Q which makes matching very sensitive and low bandwidth. Therefore, we intentionally add a resistor to introduce real part, making things low Q and match-worthy.
(One can think of ways of adding a real part to transistor output impedance without adding resistor itself and that would work too)



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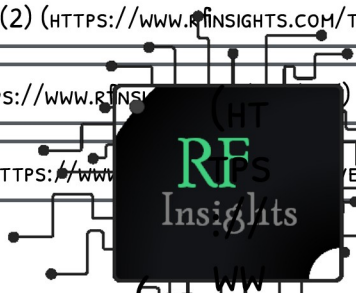
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Dedicated to my late Mother

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